Zinc

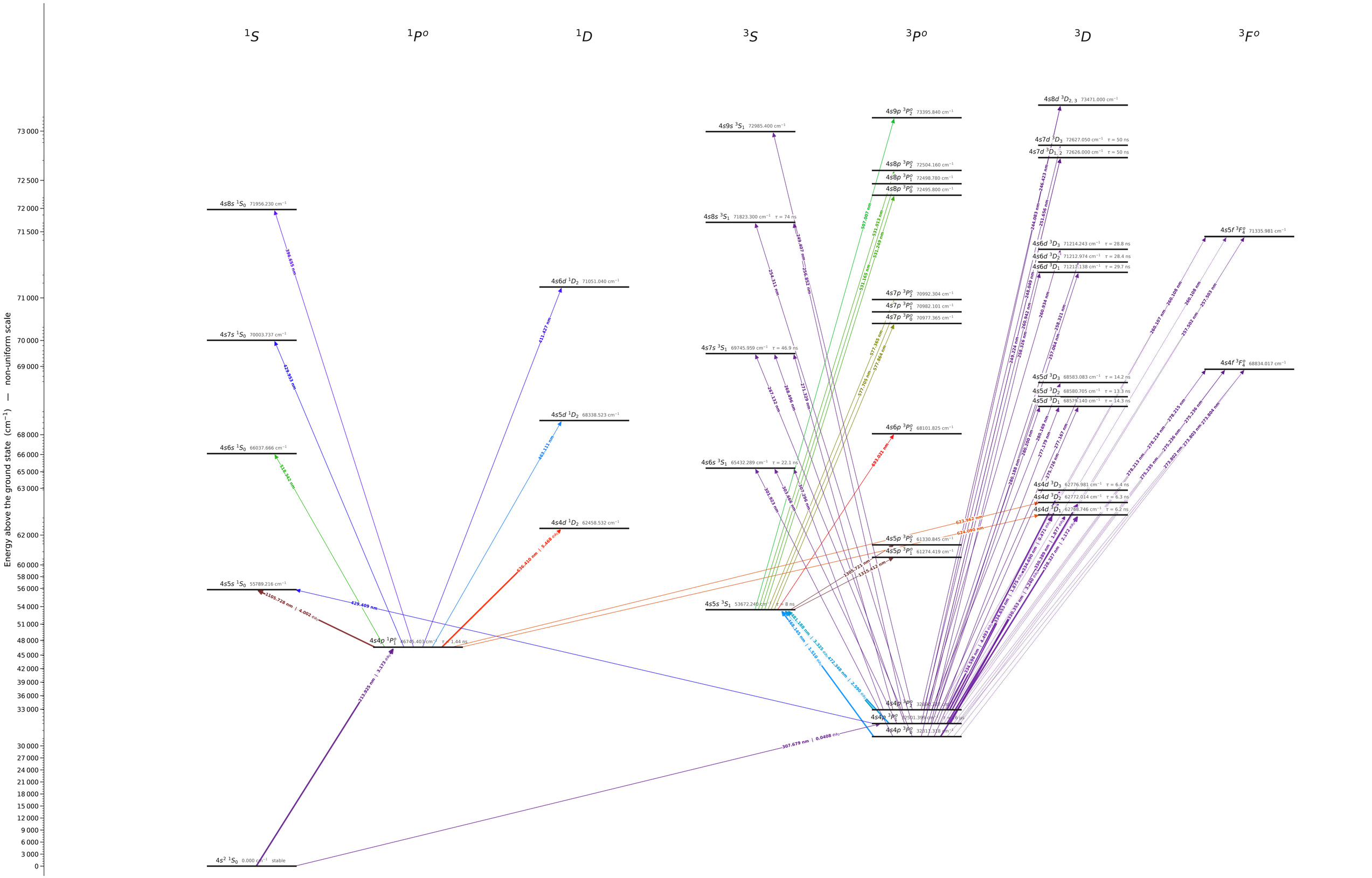
Zn (Zn I). Strongest classified transitions below 2 μm (NIST): 47 levels, 70 lines.

Zn^+ ionisation limit 75769.31 cm^{-1} = 9.39420 eV (131.980 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J\|er\|J'\rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\|\langle J\|er\|J'\rangle\|$ (ea_0)	A (s^{-1})	Γ/2π up. (MHz)	branch (%)	Use
$4s^2\ ^1S_0$ – $4s4p\ ^1P^o_1$	213.9248	213.8574	1401.39166(6)	3.173	6.944e+08	110.5	100	213.9 nm resonance line: MOT / slowing / imaging
$4s^2\ ^1S_0$ – $4s4p\ ^3P^o_1$	307.6791	307.5897	974.3674	0.041	3.850e+04	0.006121	100	307.6 nm intercombination line: narrow-line cooling
$4s4p\ ^3P^o_2$ – $4s5s\ ^3S_1$	481.1877	481.0532	623.0261	3.325	6.700e+07	19.89	53.6	481.2 nm line from the metastable 4s4p 3P2 level: colline...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

^{67}Zn hyperfine structure (I = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$4s4p\ ^3P^o_2$	608.922(3)	-18.995(10)	F=3/2: -2144.52 F=5/2: -593.73 F=7/2: +1517.56
$4s4p\ ^3P^o_2$	531.987(5)	35.806(5)	F=1/2: -3698.84 F=3/2: -2916.98 F=5/2: -1604.91 F=7/2: +250.78 F=9/2: +2668.89
$4s4p\ ^1P^o_1$	20.0(20)	10(5)	F=3/2: -63.00 F=5/2: -28.00 F=7/2: +52.50
$4s5s\ ^3S_1$	1266.5(18)	–	F=3/2: -4432.75 F=5/2: -1266.50 F=7/2: +3166.25

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$4s^2\ ^1S_0$ – $4s4p\ ^1P^o_1$	213.925	66-64	525(3)
$4s^2\ ^1S_0$ – $4s4p\ ^1P^o_1$	213.925	67-64	835(4)
$4s^2\ ^1S_0$ – $4s4p\ ^1P^o_1$	213.925	68-64	1039.8(17)
$4s^2\ ^1S_0$ – $4s4p\ ^1P^o_1$	213.925	70-64	1495(4)
$4s^2\ ^1S_0$ – $4s4p\ ^3P^o_1$	307.679	66-64	689.411(3)
$4s^2\ ^1S_0$ – $4s4p\ ^3P^o_1$	307.679	67-64	1085.933(7)
$4s^2\ ^1S_0$ – $4s4p\ ^3P^o_1$	307.679	68-64	1362.390(3)
$4s^2\ ^1S_0$ – $4s4p\ ^3P^o_1$	307.679	70-64	1958.358(7)
$4s4p\ ^3P^o_2$ – $4s5s\ ^3S_1$	481.188	64-68	-141.2(12)
$4s4p\ ^3P^o_2$ – $4s5s\ ^3S_1$	481.188	66-68	-63.6(15)
$4s4p\ ^3P^o_2$ – $4s5s\ ^3S_1$	481.188	67-68	-41.4(21)
$4s4p\ ^3P^o_2$ – $4s5s\ ^3S_1$	481.188	70-68	69.5(9)

Shift = ν(first isotope) − ν(second isotope).